

exercises pass

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1 Introduction

7. Consider a cave with a population of vampire bats, denoted $B(t)$, and a population of moths, denoted $M(t)$. A bat requires two moths within range to successfully hunt one of them; this happens at rate α , resulting in the death of one moth and the birth of one new bat. Moths enter the region at a constant rate β , and bats die at a rate γ . All three parameters α, β, γ are strictly positive.

- (a) Explain why the following system is a valid model for this problem. Justify each term.

$$\dot{B} = \alpha BM^2 - \gamma B, \quad \dot{M} = \beta - \alpha BM^2.$$

[2 marks]

- (b) Find the only stationary point for this system.

[2 marks]

- (c) Now assume $\beta = \gamma = 1$, and consider the two cases $\alpha = 1$ and $\alpha = 9$. Classify the stationary point and note whether it is stable or unstable in each case.

[4 marks]

- (d) Based on your result in part (c), how does the parameter α affect the long-term populations of bats and moths? Explain your result in terms of the model assumptions.

[2 marks]

[End of Question 7; 10 marks total]

Solution to Question 7

(a)

- αBM^2 : a successful hunt requires one bat *and* two moths to be simultaneously present, giving a rate proportional to $B \cdot M^2$. Each such event removes a moth from the population and creates a new bat, so this term appears with a positive sign in $\dot{B}$ and a negative sign in $\dot{M}$.
- $-\gamma B$: bats die at a rate proportional to the current bat population.
- β : moths enter the region at a constant background rate, independent of any other population.

(b)

Set $\dot{B} = 0$:

$$B(\alpha M^2 - \gamma) = 0 \implies B = 0 \quad \text{or} \quad M^2 = \frac{\gamma}{\alpha}.$$

If $B = 0$ then $\dot{M} = \beta > 0$, so M grows without bound; this is not a stationary point. Hence $M_0 = \sqrt{\gamma/\alpha}$.

Set $\dot{M} = 0$:

$$\beta = \alpha BM_0^2 = \alpha B \cdot \frac{\gamma}{\alpha} = \gamma B \implies B_0 = \frac{\beta}{\gamma}.$$

The unique stationary point is

$$(B_0, M_0) = \left(\frac{\beta}{\gamma}, \sqrt{\frac{\gamma}{\alpha}} \right).$$

(c)

We linearise about the stationary point (B_0, M_0) . Here

$$f(B, M) = \alpha BM^2 - \gamma B \quad \text{and} \quad g(B, M) = \beta - \alpha BM^2,$$

so

$$\frac{d}{dt} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} f_B(B_0, M_0) & f_M(B_0, M_0) \\ g_B(B_0, M_0) & g_M(B_0, M_0) \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \alpha M_0^2 - \gamma & 2\alpha B_0 M_0 \\ -\alpha M_0^2 & -2\alpha B_0 M_0 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}.$$

With $\beta = \gamma = 1$ the stationary point from part (b) is $(B_0, M_0) = (1, \alpha^{-1/2})$. Substituting, and using $\alpha M_0^2 = \alpha \cdot \alpha^{-1} = 1 = \gamma$, the matrix becomes

$$A = \begin{pmatrix} 0 & 2\alpha^{1/2} \\ -1 & -2\alpha^{1/2} \end{pmatrix}.$$

Case $\alpha = 1$: $(B_0, M_0) = (1, 1)$ and

$$A = \begin{pmatrix} 0 & 2 \\ -1 & -2 \end{pmatrix},$$

which has eigenvalues $\lambda = -1 \pm i$. Since the real part is negative, the stationary point is a *stable spiral*.

Case $\alpha = 9$: $(B_0, M_0) = (1, \frac{1}{3})$ and

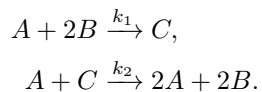
$$A = \begin{pmatrix} 0 & 6 \\ -1 & -6 \end{pmatrix},$$

which has eigenvalues $\lambda = \frac{-6 \pm \sqrt{12}}{2} = -3 \pm \sqrt{3}$. Both eigenvalues are negative (since $\sqrt{3} < 3$), so the stationary point is a *stable node*.

(d)

In both cases the stationary point is stable, so both populations converge to (B_0, M_0) in the long run. The bat population $B_0 = \beta/\gamma = 1$ is independent of α , while the moth population $M_0 = \alpha^{-1/2}$ decreases as α increases. A more efficient hunter depletes moths to a lower long-term level, but cannot affect the steady bat count, which is set by the balance of moth immigration via β and bat mortality via γ .

8. The chemicals A, Q, R have concentrations $a(t)$, $q(t)$, $r(t)$ respectively, and interact with each other according to the following set of reactions:



The concentrations obey the following ODE system:

$$\begin{aligned} \dot{a} &= -k_1ab^2 + k_2ac, \\ \dot{b} &= -2k_1ab^2 + 2k_2ac, \\ \dot{c} &= k_1ab^2 - k_2ac, \end{aligned}$$

and initially $a(0) = 5/2$, $b(0) = 3$, and $c(0) = 1/2$.

(a) Explain why $a + c = \text{constant}$ and $b + 2c = \text{constant}$. [1 mark]

(b) Hence show that the system of equations reduces to

$$\dot{a} = -4k_1a(a-1)^2 + k_2a(3-a). \quad (**)$$

[3 marks]

(c) Now let $k_1 = \frac{1}{4}$, $k_2 = 1$. Find all fixed points of equation (**) and determine their stability. [5 marks]

(d) What is the final state of the three concentrations $a(t)$, $b(t)$ and $c(t)$ as $t \rightarrow \infty$? [1 mark]

[End of Question 8; 10 marks total]

Solution to Question 8

(a) Adding the equations for $\dot{a}$ and $\dot{c}$ gives

$$\frac{d}{dt}(a + c) = \dot{a} + \dot{c} = (-k_1ab^2 + k_2ac) + (k_1ab^2 - k_2ac) = 0,$$

so $a + c$ is constant. Similarly, computing $\dot{b} + 2\dot{c}$:

$$\frac{d}{dt}(b + 2c) = \dot{b} + 2\dot{c} = (-2k_1ab^2 + 2k_2ac) + 2(k_1ab^2 - k_2ac) = 0,$$

so $b + 2c$ is also constant. □

(b) Using the initial conditions $a(0) = 5/2$, $b(0) = 3$, $c(0) = 1/2$:

$$a + c = \frac{5}{2} + \frac{1}{2} = 3, \quad b + 2c = 3 + 1 = 4.$$

Hence $c = 3 - a$ and $b = 4 - 2c = 4 - 2(3 - a) = 2(a - 1)$. Substituting into $\dot{a} = -k_1 ab^2 + k_2 ac$:

$$\dot{a} = -k_1 a [2(a - 1)]^2 + k_2 a (3 - a) = -4k_1 a (a - 1)^2 + k_2 a (3 - a). \quad (**)$$

□

(c) Setting $k_1 = \frac{1}{4}$ and $k_2 = 1$, equation (**) becomes

$$\dot{a} = -a(a-1)^2 + a(3-a) = a[-(a-1)^2 + (3-a)] = a[-a^2 + a + 2] = -a(a-2)(a+1).$$

The fixed points are the solutions of $\dot{a} = 0$, namely

$$a^* = 0, \quad a^* = 2, \quad a^* = -1.$$

To determine stability we compute $f'(a)$ where $f(a) = -a(a-2)(a+1) = -a^3 + a^2 + 2a$:

$$f'(a) = -3a^2 + 2a + 2.$$

Evaluating at each fixed point:

$$\begin{array}{ll} f'(0) = 2 > 0 & \Rightarrow a^* = 0 \text{ is } \mathbf{unstable}, \\ f'(2) = -12 + 4 + 2 = -6 < 0 & \Rightarrow a^* = 2 \text{ is } \mathbf{stable}, \\ f'(-1) = -3 - 2 + 2 = -3 < 0 & \Rightarrow a^* = -1 \text{ is } \mathbf{stable}. \end{array}$$

(d) Note that concentrations must be non-negative, and the physical domain requires $b = 2(a - 1) > 0$ and $c = 3 - a > 0$, giving $a \in (1, 3)$. The fixed points $a^* = 0$ and $a^* = -1$ both lie outside this domain. The unique physical fixed point is $a^* = 2$, which is stable on $(1, 3)$.

Since $a^* = 2$ is the unique, stable fixed point in the physical domain, and the initial condition $a(0) = 5/2 \in (1, 3)$, we have $a(t) \rightarrow 2$ as $t \rightarrow \infty$. Using the conservation laws:

$$a \rightarrow 2, \quad b = 2(a - 1) \rightarrow 2, \quad c = 3 - a \rightarrow 1.$$